

EXERCISE: INTERVAL ESTIMATION

1. Money spent on Fast Food

The data contains the amount that a sample of 15 customers spent for lunch (\$) at a fast-food restaurant:

7.42 6.29 5.83 6.50 8.34 9.51 7.10 6.80 5.90 4.89 6.50 5.52 7.90 8.30 9.60 = 106.4 $\Rightarrow \bar{x} = 7.093$

a. Find a point estimate of the population mean the amount spent for lunch (\$) at a fast-food restaurant. 7.093

b. Find the standard error of this estimate.

$$\frac{1.406}{\sqrt{15}} = 0.363$$

c. Find the margin of error for a 95% confidence interval. $t = 2.145$ / $7.093 \pm \frac{2.145 (1.406)}{\sqrt{15}} = 7.093 \pm 0.779$
= (6.315, 7.872)

d. Construct the 95% confidence interval. Interpret it in the context.

Source (Data): Berenson, M., Levine, D.M., and Krehbiel, T.C. (2012). *Basic Business Statistics:*

Concepts and Applications, 12th edition, Pearson.

2. The complaints

One of the major measures of the quality of service is the speed with which the organization responds to customer complaints. A large family held department store selling furniture. The store had the business objective of improving its response to complaints. The variable of interest was defined as the number of days between the complaints was made and when it was resolved. Data were collected from 50 complaints that were made in the past year. The data are as follows:

54 5 35 137 31 27 152 2 123 81 74 27 11 19 126 110 110 29 61 35 94 31 26 5 12 4 165
32 29 28 29 26 25 1 14 13 13 10 5 27 4 52 30 22 36 26 20 23 33 68

a. Find a point estimate of the population mean number of days between the receipt of a complaint and the resolution of the complaint.

b. Construct the 95% confidence interval. Interpret it in the context.

Source (Data): Berenson, M., Levine, D.M., and Krehbiel, T.C. (2012). *Basic Business Statistics:*

Concepts and Applications, 12th edition, Pearson.

$$\hat{p} = \frac{24}{3900}$$

3. Flu shot

In a clinical study, 3,900 subjects were vaccinated with a vaccine manufactured by growing cells in fertilized chicken eggs. Over a period of roughly 28 weeks, 24 of these subjects developed the flu.

- a. Find a point estimate of the population proportion that were vaccinated with the vaccine but still developed the flu. $\hat{p} = \frac{x}{n} = \frac{24}{3900} = 0.00615$

- b. Find the standard error of this estimate. $\sqrt{\hat{p}(1-\hat{p})} = 0.00125$

- c. Find the margin of error for a 95% confidence interval. $= (1.96)(0.00125) = 0.00245$

- d. Construct the 95% confidence interval for the population proportion. Interpret the interval. $\hat{p} \pm 0.00245$

- e. Can you conclude that fewer than 1% of all people vaccinated with the vaccine will develop the flu? $(0.00370, 0.00861) \Rightarrow 0.37\% - 0.86\% \Rightarrow \text{yes} < 1\%$

Source: Agresti, A., Franklin, C. and Klingenberg, B. (2018). Statistics: The Art and Science of Learning from Data, 4th Edition, Global Edition. Pearson

$$\begin{array}{r} \text{non-bio} \\ 306 \\ \hline 637 \end{array} \quad \begin{array}{r} \text{bio} \\ 391 \\ \hline 637 \end{array}$$

4. Basic life support knowledge

In 2015, a survey of first year university students in Brazil was conducted to determine if they knew how to activate the Mobile Emergency Attendance Service (MEAS). 306 students of non-biological subjects (n = 564) knew how to activate MEAS as compared to 391 students of biological sciences (n=637).

- a. Estimate the difference between the proportions of students of biological sciences and nonbiological subjects who know how to activate the MEAS and interpret.
- b. Construct and interpret a 95% confidence interval for the difference in proportions, explain how your interpretation reflects whether the interval contains 0.

Source: Agresti, A., Franklin, C. and Klingenberg, B. (2018). Statistics: The Art and Science of Learning from Data, 4th Edition, Global Edition. Pearson

$$\hat{p} = \hat{p} \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p}_1 - \hat{p}_2 = (\hat{p}_1 - \hat{p}_2) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

1. $\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \Rightarrow \mu = \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ | 2. $\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$

Handwritten notes:
 1. $\bar{x} - \mu$ is the difference between sample mean and population mean.
 2. $\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ is the test statistic for two independent samples.

5. Study time

Handwritten note: $\bar{x} - \mu$ is the difference between sample mean and population mean.

A graduate teaching assistant for a Statistics subject at the University of Florida collected data from students in one of her classes to investigate whether study time per week (average number of hours) differed between students in class who planned to go to graduate school and those who did not. The data were as follows:

Handwritten note: $\sigma_1 = \sigma_2$
 Graduate school: 15 7 15 10 5 5 2 3 12 16 15 37 8 14 10 18 3 25 15 5 5
 No graduate school: 6 8 15 6 5 14 10 10 12 5

No graduate school: 6 8 15 6 5 14 10 10 12 5

- Estimate the difference between the means of study time per week of two groups of students and interpret.
- Construct and interpret a 95% confidence interval for the difference means of study time per week of two groups of students.

Source: Agresti, A., Franklin, C. and Klingenberg, B. (2018). Statistics: The Art and Science of Learning from Data, 4th Edition, Global Edition. Pearson

6. Multiple myeloma

Handwritten note: $\bar{x} - \mu \Rightarrow \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
 Multiple myeloma, or blood plasma cancer, is characterized by increased blood vessel formulation in the bone marrow that is a predictive factor in survival. One treatment approach used for multiple myeloma is stem cell transplantation with the patient 's own stem cells. The data represent the bone marrow microvessel density for patients who had a complete response to the stem cell transplant (as measured by blood and urine tests). The measurement were taken immediately prior to the stem cell transplant and at the time the complete response was determined.

Patient	1	2	3	4	5	6	7
Before	158	189	202	353	416	426	441
After	284	214	101	227	290	176	290

Handwritten note: diff (d)

- a. Estimate the difference between the mean difference in bone marrow microvessel density before and after stem cell transplant.
- b. Construct and interpret a 95% confidence interval estimate of the mean difference in bone marrow microvessel density before and after stem cell transplant.

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Concepts and Applications, 12th edition, Pearson.

$$① (\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}} d \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$② (\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}} (n_1 + n_2 - 2) \sqrt{\frac{s_p^2}{1} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$